

INTERCEPTORS OF THE 1950s AND 1960s

STRATEGISTS CONTEMPLATING A POSSIBLE World War III assumed that it would start with air strikes by nuclear bombers. The prime purpose of fighter aircraft, in this respect, was to intercept enemy bombers before they could deliver their nuclear strike, with success depending above all, on speed of response. Interceptors were not expected to engage with enemy fighters, so maneuverability was not a pressing concern. They were designed for maximum acceleration and rate of climb. Rising to an altitude of around 40,000ft (12,200m) within a minute of takeoff, they would be directed on to the bombers by ground controllers and shoot them down with radar-guided air-to-air missiles. An average mission was expected to last about 10 minutes. Many of the aircraft developed for this role in the 1950s ran into problems as the new jet technology was pushed to the limits. Some Western interceptors – such as the F-106 – were dangerous if not flown by the most skillful of pilots. As usual, the Soviet Union went for simpler, cheaper machines that were easier to produce and fly.

ULTIMATE INTERCEPTOR
A Convair F-106 refuels from a KC-135 Stratotanker, while another waits its turn. The F-106 was dubbed the “Ultimate Interceptor.”



Convair F-102 Delta Dagger

Although Convair had built the world’s first delta-winged aircraft in 1948, this supersonic all-weather interceptor caused them severe embarrassment when it failed to exceed Mach 1. However, a rapid emergency redesign program, reshaping the fuselage to reduce drag, ensured that the F-102 entered service three years late in 1956.



ENGINE 17,200lb (7,802kg) thrust P&W J57-P-23A turbojet	
WINGSPAN 38ft 1in (11.6m)	LENGTH 68ft 4in (20.8m)
TOP SPEED 825mph (1,328kph) (Mach 1.25)	CREW 1
ARMAMENT 2 x nuclear or 6 x conventional Falcon guided air-to-air missiles in fuselage bay	

Convair F-106 Delta Dart

ENGINE 24,500lb (11,130kg) thrust P&W J75-P-17 turbojet	
WINGSPAN 38ft 3in (11.7m)	LENGTH 70ft 8in (21.6m)
TOP SPEED 1,265mph (2,035kph) (Mach 1.9)	CREW 1
ARMAMENT 1 x nuclear Genie unguided rocket, 4 x conventional Falcon guided air-to-air missiles in fuselage bay	

While the F-102 problems were being fixed, a second, more complete redesign was started. Originally known as F-102B, this became the F-106, as it was completely different, with a more powerful engine and the “wasp-waist” shape necessary for supersonic flight incorporated into the main design. More sophisticated fire control equipment was installed, including an automatic link to ground detection systems. The F-106 entered service in 1959.



English Electric Lightning F.1A

The Lightning was the RAF’s first supersonic fighter and the first aircraft to exceed the speed of sound in Great Britain. The Lightning could climb to over 60,000ft (18,000m) and was equipped with radar which enabled the pilot to “lock on” to the target. After 1974, Phantoms began to replace Lightnings.

ENGINE 2 x 14,430lb (6,545kg) thrust Rolls-Royce 210 turbojets	
WINGSPAN 34ft 10in (10.6m)	LENGTH 50ft (15.2m)
TOP SPEED 1,386mph (2,230kph) (Mach 2.1)	CREW 1
ARMAMENT 2 x 30mm Aden cannon in fuselage; 2 x Firestreak guided air-to-air missiles	

Gloster Javelin F(AW).9

The Javelin was the main RAF all-weather fighter from 1956–64. It was the world’s first twin-jet, delta-wing fighter, designed to intercept bombers at high altitudes and equipped with all-weather electronic equipment. After many development problems, the aircraft was finally made reliable, but by then, the Javelin was not fast enough to catch increasingly fast bombers, and was replaced by the supersonic Lightning in the early 1960s.

ENGINE 2 x 12,300lb (5,579kg) thrust Bristol Siddeley turbojet	
WINGSPAN 52ft (15.9m)	LENGTH 56ft 9in (17.2m)
TOP SPEED 701mph (1,130kph)	CREW 2
ARMAMENT 4 x 30mm Aden cannon in wings; 4 x Firestreak guided air-to-air missiles	

Thick delta wing houses engine and fuel to increase aerodynamic efficiency

Radar carried in nose



Hawker Hunter

The RAF’s main interceptor fighter from 1954 to 1960, the swept-wing Hunter replaced the Meteor and served until the introduction of the supersonic Lightning in the early 1960s. Over 1,100 were sold abroad, serving with some 20 air forces throughout the world.



ENGINE 10,000lb (4,542kg) thrust Rolls-Royce Avon turbojet	
WINGSPAN 33ft 8in (10.2m)	LENGTH 45ft 11in (14m)
TOP SPEED 715mph (1,150kph)	CREW 1
ARMAMENT 4 x 30mm Aden cannon in removable pack under fuselage; 1,000lb (454kg) bombs or rocket batteries under wings	